

NICHOLAS C. HAYEK

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EDUCATION

McGill University

B.A. Honours Mathematics and Computer Science

First-Class Honours with Distinction

Montréal, QC

08/2022 – 05/2026

- **Awards:** D. Lorne Gales Scholarship, Richard Tomlinson Undergraduate Award, Dean's Honour List.
- **Coursework:** Algorithmic Game Theory, Homological Algebra, Quantitative Risk Management, Stochastic Processes.

WORK EXPERIENCE

Department of Mathematics & Statistics, McGill

Course Assistant and TEAM Mentor

Montréal, QC (In Person)

08/2025 – 05/2026

- Run weekly tutorials for 330 students in Advanced Calculus for Engineers (vector calculus and partial differential equations).
- Review and grade homework assignments for 80 students in Honours Algebra I (group, ring, and field theory).

Indiana University, Indianapolis (IUI)

Research Internship (REU) in Mathematical Sciences

Indianapolis, IN (In Person)

06/2024 – 08/2024

- Modeled corticostriatal brain networks involved in binge alcohol drinking in rodents using a continuous-time neural network.
- Verified results from literature and analyzed model dynamics to gain insights into the biological mechanisms of addiction.
- Presented findings at the Indiana Undergraduate Math Research Conference and at the IUI Summer Research Symposium. Mentored by Prof. Alexey Kuznetsov with help from the Indiana Alcohol Research Center.

ACADEMIC PROJECTS

Director of the *McGill Undergraduate Mathematics Research Journal*

2024 – 2026

- Led the re-establishment of McGill's peer-reviewed, undergraduate math research journal.
- Recruited and managed editorial and peer review teams, fundraised, and assisted in editing and publishing.

Mathematics Transcription

2023 – 2026

- Typeset guides in linear algebra, probability, discrete math, group theory, and representation theory, among others, in LaTeX.
- Published by the McGill Society of Undergraduate Mathematics Students, with ~150 downloads per month. Example [here](#).

GARCH and ARMA Risk Computation, Hosted by You (GARCHY)

2026

- Developed an educational, [JavaScript applet](#) that locally trains, evaluates, and simulates user-specified ARMA-GARCH statistical models on custom stock portfolio data. The models can be used to forecast risk measures and volatility.
- Tooltips and interactive charts aim to educate the user on model dynamics, performance, and risk estimation.

Honours Undergraduate Thesis

2025

- Researched supersingular elliptic curve isogeny graphs (SIGs) over finite fields and their applications to cryptography.
- Implemented SIGs and their underlying algebraic structures in Python, as well as a hash function based on graph traversals, which may be used to encrypt messages. Report [here](#). Mentored by Prof. Henri Darmon.

Yet Another, Yet Another Curta Simulator (YAYACS)

2025

- Developed an online [emulator](#) of a Curta Type I mechanical calculator, with integer equation parsing and execution.

Virtual Musical Instrument Development

2024

- Developed a VST/AU virtual musical instrument based on ~350 self-recorded clips of a Steinway Model B.
- Built in C++ with 4 channels of audio. Implemented adjustable attack, decay, sustain, and release parameters and velocity-based sample selection. Users can play live or recorded music using a piano keyboard controller. Link [here](#).

ADDITIONAL INFORMATION

Computer Python, C++, C, Java, OCaml, Linux, Assembly, JavaScript (Svelte),
Data Science (Matplotlib, Pandas, NumPy, SciPy), LaTeX

Extracurricular Activities Vice President, External at the [Society of Undergraduate Mathematics Students](#) (2024-2026)
Student Researcher at the [McGill AI Lab](#) (2025)

Languages English (native)

Interests Jazz piano, reading